MODULE 1: TRADITIONAL PROGRAMMING VS ML

Introduction to Machine Learning Logic

Name: _____ Branch/Year: _____ Date: _____

> ***SCENARIO:*** *You are the Chief Engineer at a Casino Vault. A gang is using counterfeit (fake) coins to steal millions from slot machines. You need to build a system to stop them.*

Part A: The "Old-School" Scanner (Traditional Programming)

Your superstitious boss tells you: "*The fake coins always look dirty to hide the bad metal quality. Program the scanner to reject dirty coins!*"

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RULE:
IF (Coin is DIRTY) → REJECT (X)
ELSE IF (Coin is SHINY) → ACCEPT (OK)
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Task 1: Apply the rule above to these incoming coins. Check the result box.

Incoming Coin Look	Diameter	Scanner Result (Based on Rule)
Shiny Coin	24.0 mm	[ ] ACCEPT [ ] REJECT
Dirty Coin	25.0 mm	[ ] ACCEPT [ ] REJECT
Shiny Coin	25.0 mm	[ ] ACCEPT [ ] REJECT

Critical Failure: You just accepted a fake coin and rejected a real one.

Q: Why is relying only on the "Visual Look" (Shiny/Dirty) a bad way to program this scanner?

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Part B: The AI "Smart" Scanner (Machine Learning)

The old scanner failed. Now, you use Machine Learning. You **stop writing rules**. Instead, you feed the computer **Past Data** from coins the security team has already checked.

The Training Data:

Visual Look (Input 1)	Diameter (Input 2)	ACTUAL STATUS (Answer)
Shiny	25.0 mm	✅ REAL
Dirty	25.0 mm	✅ REAL
Shiny	24.2 mm	🚨 FAKE
Dirty	24.1 mm	🚨 FAKE
Shiny	23.9 mm	🚨 FAKE

Task 2 (The Test): A new, unknown coin drops into the slot. Based *only* on the pattern in the table above, is it Real or Fake?

- **New Coin:** It is extremely **Dirty**. Its Diameter is **25.0 mm**.

☐ REAL ☐ FAKE

Summary: The Difference

Comparison	Traditional Programming	Machine Learning
Input	We give the computer **Rules**. *(e.g., "If Dirty, reject it")*	We give the computer **Data**. *(e.g., "Here are 100 examples")*
Output	The computer executes the hard-coded rules.	The computer finds patterns and makes predictions according to the patterns.
Logic	Based on explicit, hard-coded rules.	Based on pattern identification in past data.

MODULE 2: CLASSIFICATION VS REGRESSION

CONTEXT: *You are given data from past exams to help understand student performance.*

Inputs:
H = Hours studied per day before the exam
S = Hours slept the night before the exam

You have 2 tasks below.

Task 1 — Predict Pass / Fail

Analyze the historical data provided in the table below.

Student	H	S	Outcome
Student 1	2.0	6.0	FAIL
Student 2	3.0	6.0	PASS
Student 3	4.0	7.0	PASS
Student 4	5.0	6.5	PASS
Student 5	6.0	7.0	PASS
Student 6	7.0	6.5	PASS
Student 7	8.0	7.5	PASS
Student 8	9.0	8.0	PASS
Student 9	10.0	7.0	FAIL

Prediction Table: Based on the patterns above, predict the outcome for these new students:

New Student	H	S	Prediction
Student A	4.5	6.8	
Student B	10.0	7.5	

Task 2 — Predict Marks

The actual numerical exam marks for the same students are now shown.

Student	H	S	Score
Student 1	2.0	6.0	48.6
Student 2	3.0	6.0	59.1
Student 3	4.0	7.0	67.8
Student 4	5.0	6.5	71.7
Student 5	6.0	7.0	73.8
Student 6	7.0	6.5	71.7
Student 7	8.0	7.5	68.4
Student 8	9.0	8.0	61.5
Student 9	10.0	7.0	49.8

Task 2.1 — Predict the Marks

New Student	H	S	Predicted Marks
Student A	4.5	6.8	_____
Student B	10.0	7.5	_____

Task 2.2 — Observe the Pattern

Use Desmos to plot: Hours Studied (H) on x-axis and Score on y-axis.

What do you notice about the relationship between studying and the resulting score?

_____

_____

This is the essence of Machine Learning: discovering the hidden mathematical equation that best fits the pattern in the data.

Think Carefully

"In the first task, the output you predicted was _____"

"In the second task, the output you predicted was _____"

Definitions

In Machine Learning, these two types of tasks have specific names:

Classification — predicting discrete or fixed categories or labels.
Examples: PASS / FAIL, Spam / Not Spam, Yes / No, Cat / Dog

Regression — predicting continuous numerical values.
Examples: marks, house price, temperature, delivery time

Relation Between Classification and Regression

In real systems, which usually happens first?

- ☐ Predict a number, then apply a cutoff
- ☐ Directly predict PASS or FAIL

Regression is done first. Classification is obtained by applying a threshold on the regression output.

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EXAMPLE:

Predicted marks → 68 (Regression)
Threshold: PASS if marks ≥ 50 (Thresholding)
Final output → PASS (Classification)
```

MODULE 3: SUPERVISED VS. UNSUPERVISED LEARNING

Part 1: Recap

Look back at Task 1 (Pass/Fail) and Task 2 (Marks):

In both cases, you were given a dataset that included both the **Inputs/Features** (Hours Studied, Sleep) and the **correct Outputs/Target/Labels** (Outcome/Score).

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SUPERVISED LEARNING
Type of machine learning where a model is trained on data having both -
inputs and outputs, to make predictions on new data.

* Classification and Regression are both types of Supervised Learning.
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Part 2: The Unknown

Now, let's explore Unsupervised Learning:

What if the data has no outputs/no labels i.e no correct answers and only input/features?

Tchibo Customer Segmentation Practical

What strategies do you recommend for the following segment of customers:

Segment 1 (Top Right) _____

Segment 2 (Top Left) _____

Segment 3 (Bottom Left) _____

Segment 4 (Bottom Right) _____

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UNSUPERVISED LEARNING
Type of machine learning where the computer finds hidden structures in
data without being given correct answers (no labels).

Examples: Customer segmentation, Clustering similar news articles,
Grouping photos by face.
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Part 3: The Hybrid & The Gamer

1. Semi-Supervised Learning

The Problem: Labeling data is expensive. Imagine you have 10,000 chest X-rays, but hiring a doctor to label them as "Healthy" or "Pneumonia" costs $100 per image (sample).

The Solution: The computer first groups similar X-rays together (Clustering). The doctor then labels just a few examples from each group. The system automatically applies these labels to the rest of the group.

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SEMI-SUPERVISED LEARNING
Type of machine learning that uses a small amount of labeled data and a
large amount of unlabeled data.

Examples: Google Photos (you name a face once, it finds it everywhere),
Medical Imaging.
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2. Reinforcement Learning

The Scenario: You are training a robot dog. You don't give it data. You just put it in a room.

- If it walks forward → You give it a **Reward** (+10 points / Treat).
- If it hits a wall → You give it a **Penalty** (-10 points / Scolding).

Question: After bumping into the wall 50 times, what will the robot eventually learn to do?

Key Insight: It doesn't wait for human-collected data. It generates its own data through rapid trial and error, often learning faster than humans.

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REINFORCEMENT LEARNING
Learning by interacting with an environment through "trial and error"
to maximize a reward.

Examples: Chess engines (AlphaGo), Self-driving cars parking, Training
video game bots.
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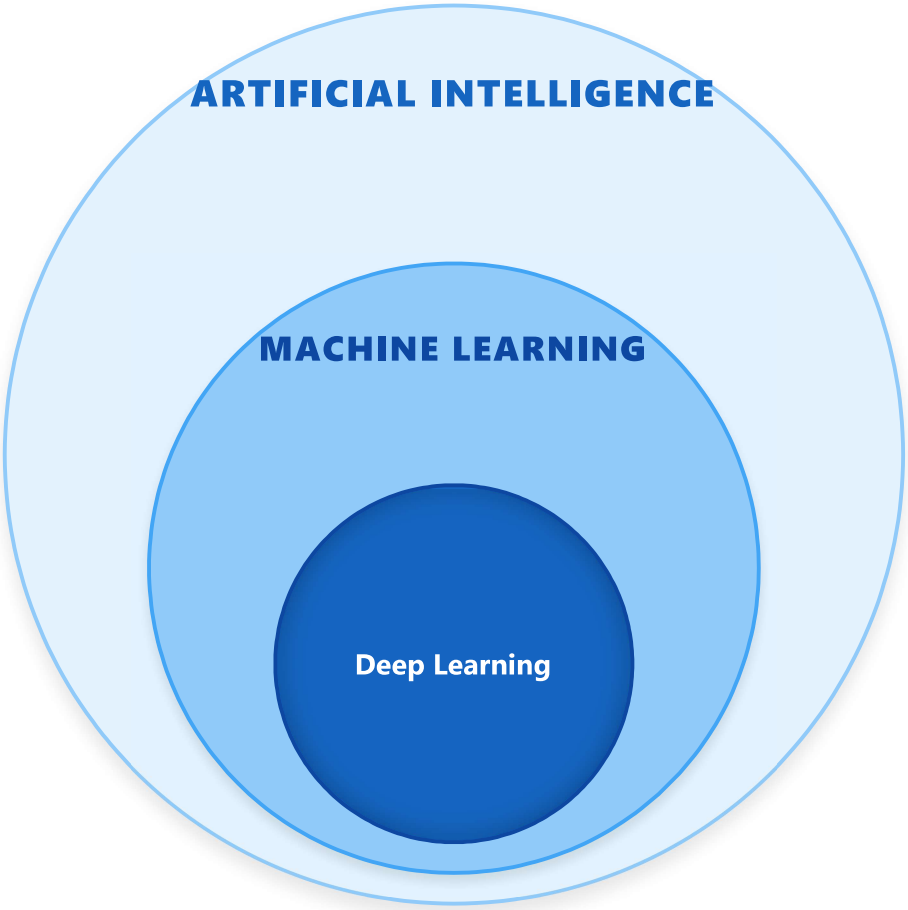
MODULE 4: AI VS. ML VS. DL

The Hierarchy of Intelligence

Visualizing the Relationship

AI, ML, and DL are not separate technologies. They are subsets of one another, similar to Russian Nesting Dolls.



Definitions

1. Artificial Intelligence (The Outer Circle)

"The Dream"

This is the broad umbrella term for any technique that enables computers to mimic human intelligence.

Definition: Any computer program that does something "smart."

Scope: This includes everything from the advanced ChatGPT down to simple programs from the 1980s that used if-else rules (Old School AI).

Key Takeaway: All Machine Learning is AI, but not all AI is Machine Learning.

2. Machine Learning (The Middle Circle)

"The Method"

This is a subset of AI. It was the shift from "telling computers what to do" to "teaching computers how to learn."

Definition: Algorithms that parse data, learn from it, and then apply what they've learned to make informed decisions.

The Difference: Instead of writing 10,000 lines of code defining what a "cat" looks like (ears, whiskers, tail), you feed the algorithm 10,000 pictures of cats and it figures out the patterns itself.

Examples: Spam filters, product recommendation engines (Netflix/Spotify).

3. Deep Learning (The Inner Circle)

"The Engine"

This is a specialized subset of Machine Learning. It is inspired by the structure of the human brain.

Definition: ML that uses Artificial Neural Networks with many layers ("deep" refers to the number of layers).

The Difference:

- **Classic ML:** You have to "hand-hold" the model by giving it specific features. For example, to predict house prices, you must explicitly provide "Area", "Bedrooms", and "Bathrooms".
- **Deep Learning:** It gets features on its own. You feed it raw data (like images), and it automatically figures out the features (edges, shapes, textures) without being told.

Examples: Image Processing (Face Recognition), Language Processing (ChatGPT/Translation), Self-Driving Cars.

Check Your Understanding

Decide whether the task can be solved with standard **Machine Learning (ML)** or requires **Deep Learning (DL)**. Please put a tick (✓) or cross (X) in the appropriate column.

Task Description	Basic ML is Sufficient	Needs Deep Learning
Recognizing Faces in a Crowd *Data: Pixel values from a camera feed (Unstructured images).*		

Task Description	Basic ML is Sufficient	Needs Deep Learning
Writing a Poem in Shakespeare's Style *Data: Complex patterns of human language and grammar (Unstructured text).*		
Predicting House Prices *Data: Square footage, number of rooms, zip code (Structured numbers).*		